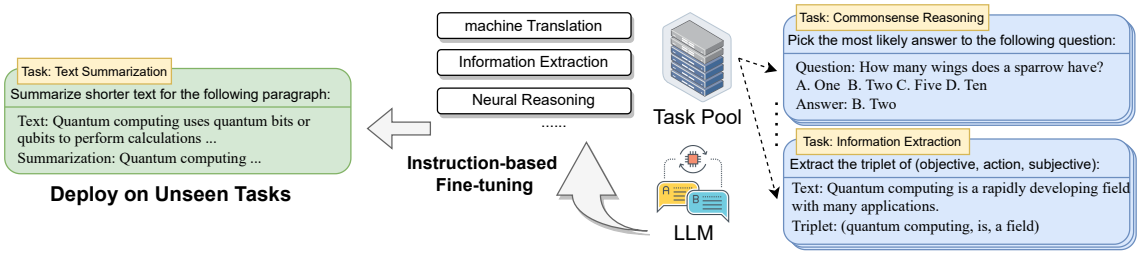

Fig. 8. The overview of fine-tuning an LLM with explicit instructions across various domains and datasets. Particularly, the LLM is fine-tuned on a collection of tasks (e.g., commonsense reasoning, information extraction, etc.) with detailed instructions, and the fine-tuned LLM is expected to obtain problem-solving skills.

reasons including but not limited to overfitting, catastrophic forgetting, and task-specific biases [163]. 2) fine-tuning LLMs is computationally expensive due to the vast parameter space and the deep model architecture. In this section, we review recent techniques on how to update the global knowledge of LLMs, which can be primarily categorized into two areas: **Instruction-based Fine-tuning** and **Partial Knowledge Update** to address both challenges, respectively.

5.2.1 Instruction-based Knowledge Update. Instruction-based Knowledge Update refers to updating an LLM's parametric knowledge by fine-tuning LLMs on a diverse set of tasks with explicit instructions or prompts, which is conceptually the same as Instruct Learning introduced in [111]. An illustration of fine-tuning an LLM with instructions is provided in Figure 8, where an LLM is fine-tuned on a collection of tasks across the whole NLP application domain, and the LLM is deployed on the held-out and unseen tasks. Wei et al. f[165] provided the very first attempt to fine-tune LLMs based on a collection of datasets described via instructions. Empirically, effective instructions can substantially improve zero-shot performance on unseen tasks. The instruction-tuned language model FLAN is fine-tuned on a 137B LLM over 60 NLP datasets using natural language instruction templates. The study shows that FLAN outperforms its unmodified counterpart and even surpasses both zero-shot and few-shot 175B GPT-3 on most unseen tasks. Subsequently, in recent works by [23, 56, 98], explicit instructions have been employed to fine-tune LLMs, with emphasis placed on (1) expanding the number of tasks, (2) enlarging the model's size, and (3) fine-tuning on chain-of-thought data. As a result, the fine-tuned LLM attains state-of-the-art performance on numerous benchmarks in both zero/few-shot NLP tasks.

Fine-tuning with Human Instructions. Fine-tuning with human instructions aims to guide LLMs towards generating safer, truthful, less toxic content in line with user intentions. Most LLMs utilize autoregressive approaches, making the generated content largely influenced by the training corpus distribution and less controllable. Reinforcement learning from human feedback (RLHF) is a notable technique for aligning LLM content with human needs [21]. In RLHF: 1) LLMs create multiple content options for a prompt, ranked by humans for quality, relevance, and desired output alignment; 2) an external reward model assigns scores to content based on rankings, capturing evaluator preferences; 3) model policy is updated using reinforcement learning techniques to maximize expected reward, fine-tuning the model to better align with human preferences; 4) this process of content generation, ranking, reward modeling, and policy optimization repeats in iterations, with the model continually learning from human feedback. Existing methods successfully apply RLHF to fine-tune LLMs on complex reasoning tasks using human instructions [113, 156].

Potential Limitations of Instruction-based Knowledge Update. Knowledge updates based on explicit instructions tend to perform well on Natural Language Understanding tasks but are limited to simpler instructions and struggle with tasks diverging from evaluation sets. Improving adaptability to diverse tasks often incurs catastrophic forgetting. A crucial

question is extending model knowledge and abilities without causing such forgetting. Recently, Huang et al. proposed a method that uses a pre-trained LLM to generate high-confidence, rationale-augmented answers for unlabeled questions, improving general reasoning without ground truth labels or explicit instructions [55]. Additionally, Scialom et al. are expanding LLM knowledge and abilities without forgetting previous skills by fine-tuning LLMs across various tasks and introducing an approach to counter catastrophic forgetting with Continual Learning via Rehearsal [138, 141].

5.2.2 Partial Knowledge Update. Other than leveraging task-specific instructions to fine-tune LLMs, a number of approaches emerge to conduct LLM fine-tuning by updating/editing a part of LLM parameters that link to specific knowledge without leveraging external guidance. Suppose $f_\Theta(\cdot)$ denotes the function of LLM parametrized with the set of parameters Θ and $\theta \in \Theta$ is the single parameter in Θ. Updating the inner knowledge of $f_\Theta(\cdot)$ based on a collection of training data D is denoted as:

$$\tilde{\Theta} = \Theta + \nabla f_\Theta(D) \odot T, \qquad T^{(i)} = \begin{cases} 1, & \text{if } \theta^{(i)} \in \Theta_T \\ 0, & \text{if } \theta^{(i)} \notin \Theta_T \end{cases} \tag{1}$$

where T denotes the mask vector and $T^{(i)} \in T$ denote the i-th element of T. The mask controls the amount of LLM's inner knowledge to be updated in each fine-tuning iteration, where we use $\Theta_T \subseteq \Theta$ to denote the parameters that need to be updated in Θ. In the conventional setting of fine-tuning pre-trained language models [50, 130, 189], $|\Theta| = |\Theta_T|$. However, updating all of the parameters is computationally prohibited and resource-consuming in the context of LLM. Empirically, $|\Theta| \gg |\Theta_T|$, which refers to the modification of only a small number of parameters. Existing parameter-efficient knowledge update can be categorized into three streams: i.e., **Knowledge Editing** aims at directly locating and updating a small subset of parameters in an LLM; **Gradient Masking** aims at masking out the gradients of non-relative parameters during the fine-tuning; and **Knowledge Distillation** focuses on obtaining a child model with domain-specific knowledge from LLMs.

Knowledge Editing. Recent research has seen success in updating LLMs with new memories to replace outdated information or add specialized domain knowledge. For instance, improving the ability to update an outdated prediction like "Boris Johnson is Prime Minister of the UK" can enhance an LLM's reliability and generalization. Various methods have been proposed to locate and edit an LLM's parametric knowledge [25, 28, 48, 95, 96]. De Cao et al. proposed a hyper-network trained to update LLM parameters with a single fact needing modification, avoiding fine-tuning to prevent performance degeneration [28]. However, later works found that hyper-network-based editing falters as the LLM scales up, proposing retrieval-based methods to store edits in explicit memory and reason over them to adjust LLM predictions [103, 104]. Other methods focus on localizing and understanding LLM internal mechanisms. Notable works identify crucial neuron activations for LLM factual predictions through attention mechanisms and causal interventions, successfully updating domain facts [25, 95, 96]. A recent method is proposed learning a map from textual queries to fact encodings in an LLM's internal representation, using these encodings as knowledge editors and probes [48].

Gradient Masking. Gradient masking is a technique used to selectively update specific parts of an LLM during the fine-tuning process. The main goal is to reduce computational overhead and potentially mitigate issues such as catastrophic forgetting or overfitting, particularly when adapting pre-trained models to smaller or specialized datasets. Gradient masking involves modifying the gradients during back-propagation by applying a masking function (Equation (1)). This function determines which parts of the model will be updated, effectively *masking* the gradients for certain

parameters and keeping them unchanged. The choice of parameters to mask can be based on various criteria, such as their relevance to the task, importance in the model, or contribution to the overall loss.

Earlier attempts [62, 178] have been made to efficiently fine-tune relatively small language models by utilizing various regularization techniques, their methods cannot easily adapt to fine-tuning LLMs. This is primarily due to the substantially larger amounts of data and computational resources required to train LLMs effectively, which can be several orders of magnitude more than what is needed for smaller language models. To add gradient masks to LLMs, CHILD-TUNING [171] utilizes the downstream task data to detect the most task-related parameters as the child network and freezes the parameters in non-child network to their pre-trained weights. Moreover, Zhang et al. [181] propose a Dynamic Parameter Selection algorithm for efficiently fine-tuning LLMs, which adaptively selects a more promising sub-network to perform staging updates based on gradients of back-propagation, which brings great improvement in domain-specific downstream tasks under low-resource scenarios.

Knowledge Distillation. While most works on LLM self-knowledge update focus on task-specific instructions and parameter efficiency, a promising area of research explores distilling domain-specific knowledge from LLMs into smaller networks to reduce inference latency and enhance domain-specific task solving ability. Muhamed et al. compressed a 1.5 billion parameter LLM to a 70 million parameter model for Click-through-rate prediction, introducing twin-structured BERT-like encoders and a fusion layer for a cross-architecture distillation from a single LLM, resulting in superior performance in both online and offline settings [106]. Similarly, [4, 94, 154] employ a knowledge distillation module for LLM fine-tuning, achieving faster convergence and better resource utilization. This module leverages pre-trained parameters for quick convergence and trains a small subset of parameters to address model over-parameterization. Furthermore, [51, 142] distill the step-by-step chain-of-thought reasoning abilities of larger models into smaller models.

5.2.3 Open Challenges. Fine-tuning LLMs with the latest data ensures that they provide relevant and accurate information, especially in domains where rapid changes occur, such as technology, medicine, and current events. Furthermore, we have observed different applications or users may have unique requirements or preferences. However, fine-tuning the large-scale LLMs also poses several open challenges:

(1) *Compliance with regulations*: In most cases, updating and fine-tuning LLMs are necessary to ensure compliance with specific regulations or guidelines, such as data protection laws or industry-specific requirements. The so-called *LLM alignment* can be accomplished during the fine-tuning phase.
(2) *Computational resources*: Fine-tuning or updating inner knowledge of LLMs necessitates access to high-performance GPUs or specialized hardware, which can be expensive and difficult to obtain, particularly for individual researchers or smaller organizations. Pursuing fine-tuning efficiency is still a practical yet essential problem.

6 APPLICATIONS OF LLM DOMAIN SPECIALIZATION

In this survey paper, we explore the applications of LLMs across a range of domain-specific tasks in social sciences (e.g., education, finance, law), natural sciences (e.g., biomedicine, earth science), and formal sciences (e.g., human-computer interaction, software engineering, and cyber security). To achieve domain specialization for LLMs in these diverse fields, readers can employ various techniques, such as external augmentation, instruction crafting, and knowledge update. These approaches can help tailor LLMs to specific tasks and challenges in each domain, enabling more accurate, relevant, and effective applications. Although each domain has its unique challenges and requirements, several common applications of specialized LLMs are shared across these fields: